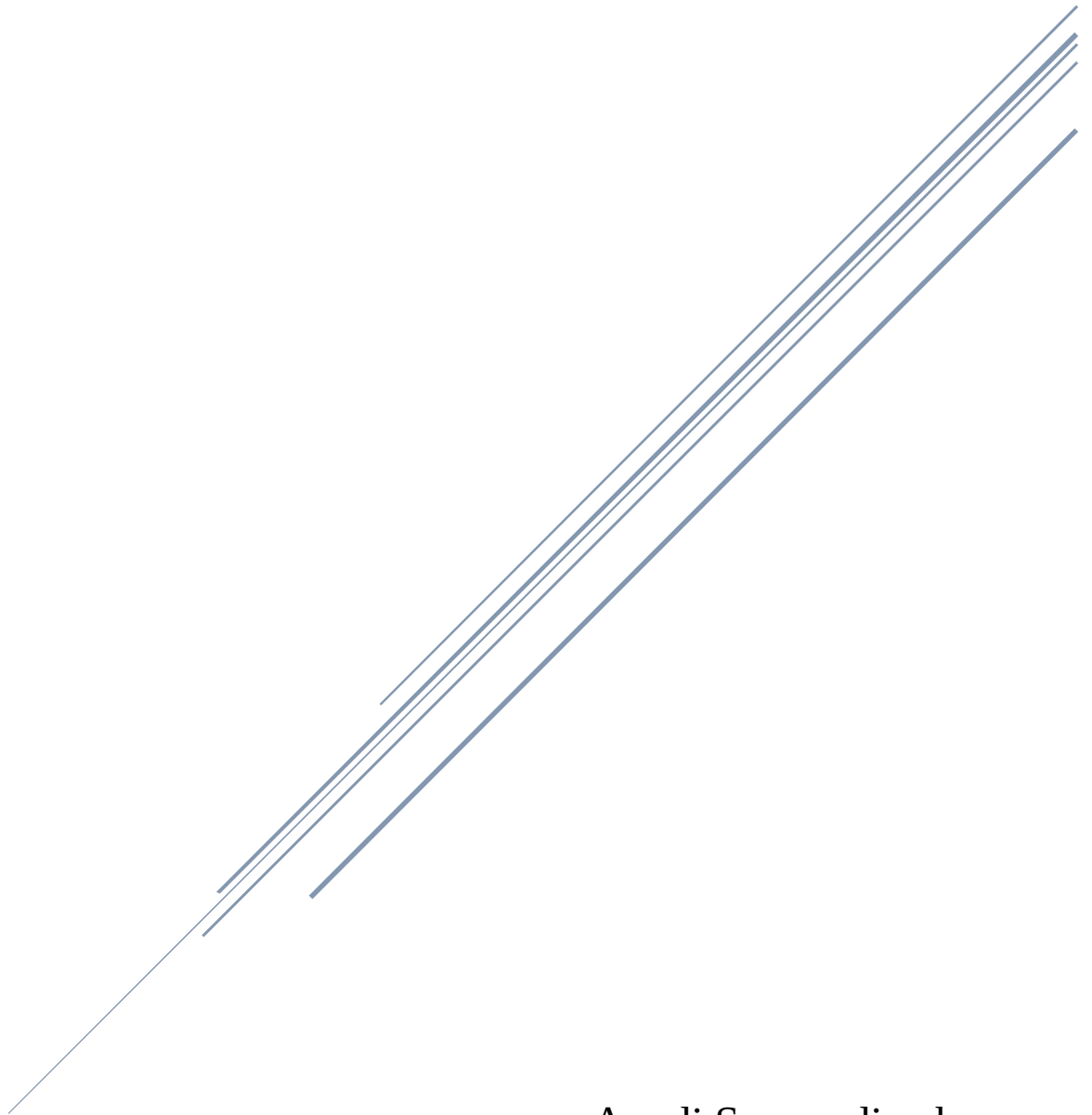


Password Strength Checker

DecodeLabs Cyber Security Internship – Project 1



Anudi Samaradiwakara
Cybersecurity

Project Overview

The Password Strength Checker is a cybersecurity project designed to analyze the strength of a user's password based on several important security factors. The purpose of this project is to help users create stronger passwords and understand the importance of password security in protecting sensitive information.

This project focuses on fundamental cybersecurity concepts such as data validation, password complexity, string handling, and conditional logic.

Project Objectives

The primary objectives of this project are:

- Evaluate the strength of a password.
- Identify weak password patterns.
- Encourage users to create stronger passwords.
- Provide security recommendations.
- Demonstrate the use of basic cybersecurity principles.

Functional Requirements

The system should perform the following tasks:

- Accept a password from the user.
- Check the overall length of the password.
- Determine whether uppercase letters are present.
- Determine whether lowercase letters are present.
- Detect numerical characters.
- Detect special symbols.
- Classify the password as weak, medium, or strong.
- Display suggestions for improving password security.

Password Evaluation Criteria

Weak Password

A password is considered weak when:

- It contains fewer than eight characters.
- It lacks uppercase letters.
- It lacks numbers.
- It lacks special characters.

Example:

ict

Medium Password

A password is considered medium when:

- It contains at least eight characters.
- It contains uppercase letters.
- It contains lowercase letters.
- It contains numbers.
- It may not contain special characters.

Example:

ict@123

Strong Password

A password is considered strong when:

- It contains at least eight characters.
- It contains uppercase letters.
- It contains lowercase letters.
- It contains numerical characters.
- It contains special symbols.

Example:

iCt@123#456...

System Workflow

Step 1

The user enters a password.

Step 2

The system analyzes the password based on predefined security rules.

Step 3

The application checks the following components:

- Password length
- Uppercase letters
- Lowercase letters
- Numbers
- Symbols

Step 4

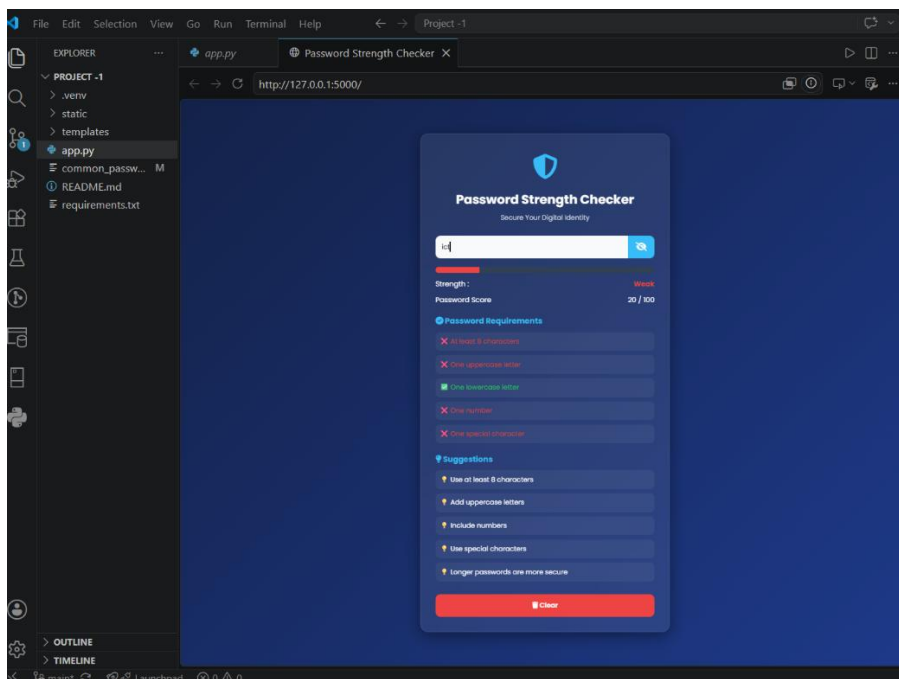
The system calculates the password strength.

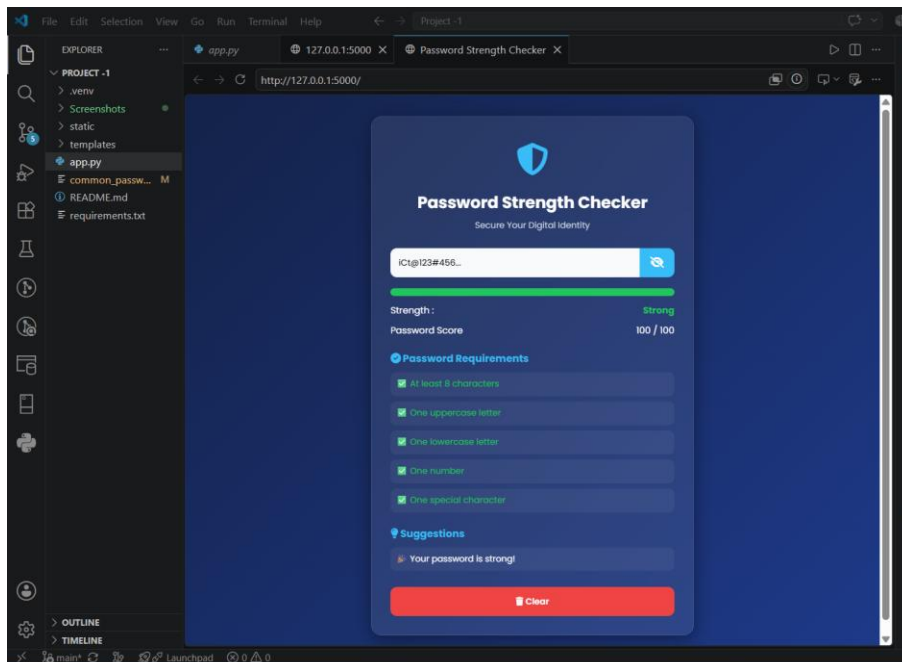
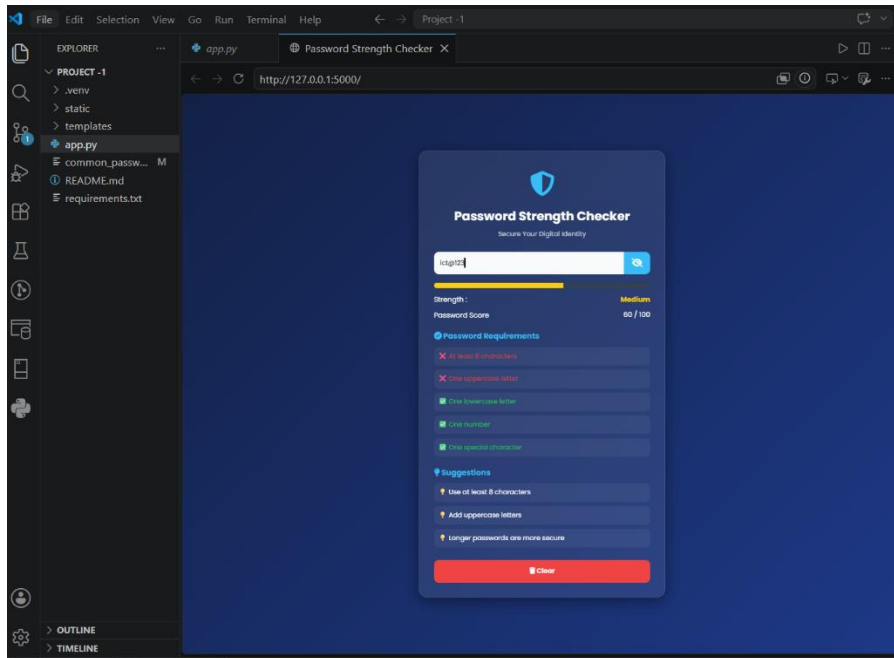
Step 5

The final result is displayed to the user.

```
File Edit Selection View Go Run Terminal Help Project-1
EXPLORER
PROJECT-1
  .venv
  static
  templates
  app.py
  common_passwords.txt
  README.md
  requirements.txt

app.py
1 from flask import Flask, render_template, request, jsonify
2 import string
3
4 app = Flask(__name__)
5
6
7 # Load common passwords
8 def load_common_passwords():
9     try:
10         with open("common_passwords.txt", "r") as file:
11             return file.read().splitlines()
12     except FileNotFoundError:
13         return []
14
15 common_passwords = load_common_passwords()
16
17
18 def check_password_strength(password):
19     score = 0
20     suggestions = []
21
22     # Check common passwords
23     if password.lower() in common_passwords:
24         return {
25             "strength": "Very Weak",
26             "score": 0,
27             "suggestions": [
28                 "This password is commonly used.",
29                 "Choose a unique password."
30             ]
31         }
32
33     # Length check
34     if len(password) >= 8:
```





User Interface Features

The graphical interface provides the following features:

- Password input field
- Show or hide password option
- Password strength indicator
- Security analysis section
- Suggestions section
- Clear button
- User-friendly layout

Technologies Used

- Python
- Tkinter
- Visual Studio Code

Advantages of the System

- Improves password security awareness.
- Encourages better password creation practices.
- Provides instant feedback.
- Offers a simple and interactive user interface.
- Enhances fundamental cybersecurity skills.

Future Enhancements

The following improvements can be added in future versions:

- Common password detection
- Password history tracking
- Password generation functionality
- Dark mode support
- Database integration
- Localhost deployment using Flask

Conclusion

The Password Strength Checker is a simple yet effective cybersecurity application that demonstrates the importance of password security. Through password analysis and security recommendations, users can understand how to create stronger passwords and reduce the risk of unauthorized access.

This project also provides practical experience in Python programming, graphical user interface development, cybersecurity concepts, and problem-solving techniques.